



United International University

Department of Computer Science and Engineering

SOC 2101: Society, Environment and Engineering Ethics

Final Examination — Summer 2025

Total Marks: 25 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. A company named *NeuroLink Systems* develops advanced brain-computer interface (BCI) devices that allow users with paralysis to control computers and robotic arms through neural signals. During clinical testing, the engineering team notices occasional system malfunctions where unintended electrical impulses are sent to the user's neural implant, causing mild but unexpected muscle spasms.

Despite these findings, the company's leadership decides to accelerate product launch to gain a market advantage, arguing that the risks are "minor and acceptable." The engineers are told not to document these anomalies in the final safety report. Meanwhile, a data-analytics subcontractor working with *NeuroLink Systems* is discovered to be using participants' neural activity data to train its internal AI models, without obtaining explicit consent.

Dr. Rafiq, a senior IEEE member and lead engineer on the project, faces an ethical dilemma: report these issues and risk his job, or remain silent to protect his position and the company's public image.

- (a) Which IEEE ethical principle(s) were violated and which were followed? [3]
- (b) What should Dr. Rafiq and the company do to act ethically and responsibly? [2]

2. A city police department deploys a **Predictive Policing System** that uses Machine Learning to forecast **crime hotspots** and identify individuals at risk of reoffending. Trained on ten years of arrest and neighborhood data, the system is praised for improving patrol efficiency.

Over time, however, activists notice that it consistently labels low-income and minority neighborhoods as "high-risk," resulting in heavier policing and more arrests - a concerning issue.

One day, Jamal, a 19-year-old college student with no criminal record, is stopped for questioning because the AI flagged his neighborhood. Feeling targeted, he files a complaint, claiming the system is **racially biased** and **lacks transparency**. When asked, police officials admit they cannot explain the model's decision process due to its proprietary design..

Identify the type of bias(es) in the given scenario. How should the police department and city officials ethically address the concerns about **Algorithmic Bias, Transparency, and Accountability** in predictive policing systems? Also, how can the system be made better in terms of accurately identifying the criminals? And how can the decision-making process be made more reliable? [5]

3. A group of university students initiated a research project on **childhood abuse and trauma** to examine their impact on individuals' **later personal development** and **professional performance**. Due to limited funding and the accessibility of online data, they collected information from **social media forums** where people shared their experiences of childhood trauma. Some users had even posted **psychiatric prescriptions** while discussing their treatment, thereby revealing details about their mental health conditions. The students downloaded these materials and used them for analysis. Additionally, some students accessed the **personal profiles** of forum participants to gather further background information.

When the course teacher reviewed their research, he criticized the students for failing to follow appropriate ethical and methodological standards. The students had included detailed data, analysis results, and personal information in their report, believing that greater detail would improve their grades.

What are the ethical concerns associated with the students' research on childhood abuse and trauma? How could this study have been conducted correctly to ensure ethical compliance, data privacy, and research integrity? [5]

4. A research team from a health-tech startup in Dhaka is conducting a study on stress levels among university students using a mobile app. The app collects heart rate, sleep patterns, and daily mood entries. During data collection, students are not given detailed consent forms—only a short note saying "By using this app, you agree to share your data." Later, the company shares aggregated results publicly but also accidentally leaks some identifiable records (like student names and heart rates).

- (a) What ethical principle was violated when students were not provided with a detailed consent form? [1]
- (b) What should have been included in a proper informed consent document in this study? [2]
- (c) When the company shared identifiable health data publicly, which ethical aspect was compromised? [2]

5. In 2025, major tech companies and research institutions significantly expanded the use of **Large Language Models (LLMs)** such as GPT, Gemini, and Llama in sectors like healthcare, education, finance, and customer service. These models run on massive data centers requiring thousands of high-performance GPUs and continuous energy consumption. Supporters argue that LLMs improve **efficiency, reduce human labor, enable personalized services**, and accelerate scientific discovery. Additionally, AI-powered optimization can support sustainability efforts, such as improving energy grids, predicting climate patterns, and reducing waste in supply chains. However, critics raise concerns about the **environmental footprint** of developing and deploying LLMs. Training a single large model can consume millions of kWh of electricity and generate significant CO₂ emissions. The production of GPUs relies on rare earth metals and **creates e-waste** when hardware becomes obsolete. Furthermore, the constant scaling of model sizes increases infrastructure demands and widens the digital divide between wealthy and developing regions.

Analyze the environmental impacts of LLM-based computer technologies. Discuss both **positive and negative effects**. Propose **strategies to minimize environmental harm** while maximizing societal and sustainability benefits. [5]

Annexure: IEEE Codes of Ethics

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members, and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree –

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons and to not engage in acts of discrimination based on race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.